MARIA FEBUS

HEALTHCARE DATA SCIENTIST | AI & PROJECT LEADERSHIP

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SUMMARY

Healthcare analytics professional with 15+ years in pharmaceutical R&D and supply chain, evolved from IT project leadership into data science and AI. Combines hands-on expertise in Python, machine learning, and statistical modeling with deep experience in GxP-regulated environments and a proven record leading cross-functional initiatives. Translates complex analyses into decision-ready insights for executive and technical stakeholders, with a growing portfolio of independent healthcare AI projects.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Scikit-learn), SQL, TensorFlow
- **Machine Learning & Deep Learning:** Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, Convolutional Neural Networks (CNN), LLMs / RAG / Agentic AI
- **Analytics:** Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Model Evaluation
- **Simulation & Modeling:** Discrete Event Simulation (DES), Agent-Based Modeling (ABM) using AnyLogic
- **Data Visualization:** Matplotlib, Seaborn, Altair, Power BI, Qlik Sense
- **Platforms:** MySQL, PostgreSQL, Informatica, Azure, GitHub
- **Project & Regulated Systems:** SDLC, Agile/Scrum, Computer System Validation (CSV), GxP
- **Lean Six Sigma:** DMAIC – SIPOC, Voice of Customer (VoC), Critical-to-Quality (CTQs)

SELECTED DATA SCIENCE & AI PROJECTS

Predicting Myocardial Infarction and Identifying High-Risk Profiles — COMPLETED · JUNE 2026

Supervised classification combined with unsupervised clustering to identify individuals at risk of myocardial infarction and surface distinct high-risk profiles for clinicians and care coordinators seeking to prioritize preventive interventions; independently extending into an agentic AI clinical decision-support system (CardioAssist).

Breast Cancer Detection — CNN Classifier — COMPLETED · MARCH 2026

Custom CNN trained on the IDC histopathology dataset to classify benign vs. malignant tissue, optimized for recall to minimize missed diagnoses — a scalable AI-assisted triage tool for pathologists and oncology teams exploring AI-assisted diagnostic support.

Diabetes Prevalence & Chronic Disease Correlation Analysis — COMPLETED · FEBRUARY 2026

Cross-state correlation analysis combining diabetes prevalence with comorbidity indicators across all 50 US states, informing payer risk modeling, chronic care program design, and population health prioritization for public health agencies and policymakers.

Predicting Drug Shortages Before They Happen — EXPECTED · JULY 2026

Classification model (XGBoost) trained on FDA shortage data, with features engineered from manufacturer concentration, drug class, and supply chain signals, giving pharmaceutical manufacturers, distributors, and health systems an early-warning capability to support proactive inventory planning and supply chain resilience.

Medication Adherence Segmentation — EXPECTED · AUGUST 2026 · CAPSTONE

Combined unsupervised clustering and supervised classification to discover patient medication adherence archetypes and assign new patients to the segment most predictive of their behavior, supporting tailored interventions for health plans, patient support programs, and care management organizations.

PROFESSIONAL EXPERIENCE – SELECTED PROJECTS

Senior IT Manager, MedTech ERP Portfolio — *Johnson & Johnson, MedTech* — 2022 – 2024

- Led technology governance across a global, multi-workstream ERP transformation spanning Make and Order to Cash capabilities — managing ~$150M in technology investments and 100+ employees/contractors — and established executive-level reporting on financials and delivery status that closed a prior visibility gap for leadership.
- Developed the MedTech Legacy ERP Decommission Roadmap, performing impact analyses that identified $59M in projected decommissioning savings.

- Informed enterprise investment decisions by delivering portfolio updates, financial analyses, and strategic recommendations for semiannual business planning cycles.

IT Project Manager — Data Management & Analytics *— Johnson & Johnson, Janssen R&D* **2015 – 2022**

- Led development of an automated Peripheral Blood Mononuclear Cell (PBMC) Sample Management System, replacing manual, Excel-based CRO tracking with end-to-end chain-of-custody visibility and CRO performance metrics — strengthening data integrity and earning the 2022 J&J Innovation Leadership Award.
- Directed data integration and analytics support for COVID-19 vaccine programs aligned with Emergency Use Authorization (EUA) submissions.
- Led creation of a Clinical Operations IT Portfolio Governance Model — closing a critical visibility gap for senior leaders into a $33M IT portfolio — through executive dashboards, structured project reviews, and proactive risk mitigation.
- Led implementation of Enterprise Data Lake, consolidating and harmonizing fragmented clinical ops and R&D data to enable advanced analytics across Clinical Operations.

IT Project Manager *— Johnson & Johnson, Janssen Supply Chain* **2010 – 2015**

- Spearheaded design and implementation of IT New Product Introduction (NPI) processes supporting 24-hour Rapid U.S. Launches, confirming distribution-center inventory readiness within 24 hours of FDA approval.
- Led SAP configuration supporting a direct-to-provider distribution model, enabling compliant dispense to 350 U.S. physicians and optimizing end-to-end supply chain workflows.

ADDITIONAL PROJECTS

Johnson & Johnson, Janssen Supply Chain

- Led launch of the NPMonitor reporting portal in a dual Project Manager/Scrum Master role, replacing fragmented manual status updates with a single cross-functional view of product status from drug substance through drug product manufacturing.
- Implemented an SAP Master Data Management platform consolidating pharma product master data from 5 ERP systems, improving data quality and accessibility for critical business processes and strategic projects.
- Led development of a discrete-event simulation model for the EPO/EPREX bio-manufacturing process, enabling dynamic scenario evaluation and supporting strategic recommendations for plant expansion and resource allocation — contributing to an innovation recognized with the J&J IT IMAGE Award.

EDUCATION

- Master of Applied Data Science and AI (MADS) – Expected August 2026, University of Michigan
- Post-Graduate Certificate, Agentic AI — Expected October 2026, Johns Hopkins University
- Post-Graduate Certificate, AI in Healthcare — Completed 2025, MIT xPRO
- Post-Graduate Certificate, Applied Data Science — Completed 2025, MIT xPRO
- Bachelor of Business Administration, Computer Information Systems — Summa Cum Laude, Baruch College (CUNY)

CERTIFICATIONS

- Certified Project Management for AI (PMI-CPMAI) – Expected December 2026
- Project Management Professional (PMP) 2009 – 2024
- Certified Scrum Master (CSM) 2021 – 2024

AWARDS

- Johnson & Johnson Innovation Leadership Award | Johnson & Johnson IT IMAGE Award